AsqTad Links

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The AsqTad smear [1, 2, 3] is

$$V_\mu = c_1 U_\mu + c_N U_\mu^{(N)} + \sum_{\nu\neq\mu} \left[c_3 U_{\mu\nu}^{(3)} + \sum_{\substack{\rho\neq\mu \\ \rho\neq\nu}} \left(c_5 U_{\mu\nu\rho}^{(5)} + \sum_{\substack{\sigma\neq\mu \\ \sigma\neq\nu \\ \sigma\neq\rho}} c_7 U_{\mu\nu\rho\sigma}^{(7)} \right) + c_L U_{\mu\nu}^{(L)} \right], \tag{1}$$

where summations run over both forward and backward directions, and the link constructs are given by

$$\begin{aligned} U_\mu^{(N)}(x) &= U_\mu(x) U_\mu(x+a\hat{\mu}) U_\mu(x+2a\hat{\mu}), \\ U_{\mu\nu}^{(3)}(x) &= U_\nu(x) U_\mu(x+a\hat{\nu}) U_\nu^\dagger(x+a\hat{\mu}), \\ U_{\mu\nu\rho}^{(5)}(x) &= U_\nu(x) U_{\mu\rho}^{(3)}(x+a\hat{\nu}) U_\nu^\dagger(x+a\hat{\mu}), \\ U_{\mu\nu\rho\sigma}^{(7)}(x) &= U_\nu(x) U_{\mu\rho\sigma}^{(5)}(x+a\hat{\nu}) U_\nu^\dagger(x+a\hat{\mu}), \\ U_{\mu\nu}^{(L)}(x) &= U_\nu(x) U_{\mu\nu}^{(3)}(x+a\hat{\nu}) U_\nu^\dagger(x+a\hat{\mu}). \end{aligned} \tag{2}$$

The first 3-link construct represents a third-nearest-neighbor interaction called the *Naik* term. The last 5-link construct $U_{\mu\nu}^{(L)}$, a 5-link staple that is restricted to a plane, is called the *LePage* term. The coefficients are typically

$$c_1 = \frac{1}{8}, \quad c_3 = \frac{1}{16u_0^2}, \quad c_5 = \frac{1}{64u_0^4}, \quad c_7 = \frac{1}{384u_0^6}, \tag{3}$$

$$c_L = -\frac{1}{16u_0^4}, \quad c_N = -\frac{1}{24u_0^2}, \tag{4}$$

where u_0 is the *tadpole factor*. When $c_L = c_{\text{Naik}} = 0$ and $u_0 = 1$, this smear reduces to Fat7.

References

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